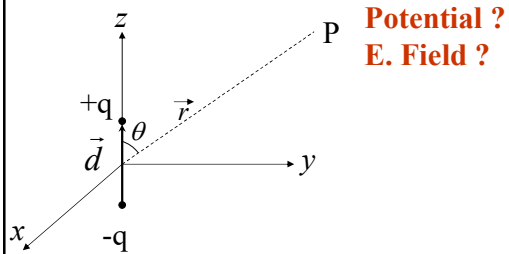
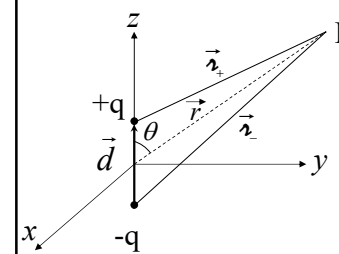


Simplest possible charge system



1

Simplest possible charge system Potential



$$r_+ = (r^2 + (d/2)^2 - rd \cos \theta)^{1/2}$$

$$r_- = (r^2 + (d/2)^2 - rd \cos(\pi - \theta))^{1/2}$$

$$= (r^2 + (d/2)^2 + rd \cos \theta)^{1/2}$$

$$r_{\pm} = (r^2 + (d/2)^2 \mp rd \cos \theta)^{1/2}$$

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{r_+} - \frac{q}{r_-} \right)$$

$$\frac{1}{r_{\pm}} = \frac{1}{r} \left(1 \mp \frac{d}{r} \cos \theta + \frac{d^2}{4r^2} \right)^{-1/2}$$

$$\rightarrow \frac{1}{r_{\pm}} = \frac{1}{r} \left(1 \mp \frac{d}{r} \cos \theta \right)^{-1/2} \quad r \gg d$$

$$\approx \frac{1}{r} \left(1 \pm \frac{d}{2r} \cos \theta \right)$$

2

Simplest possible charge system Potential

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{r_+} - \frac{q}{r_-} \right)$$

$$\frac{1}{r_{\pm}} \approx \frac{1}{r} \left(1 \pm \frac{d}{2r} \cos \theta \right)$$

$$\rightarrow \frac{1}{r_+} - \frac{1}{r_-} = \frac{1}{r} \left(1 + \frac{d}{2r} \cos \theta - 1 + \frac{d}{2r} \cos \theta \right)$$

$$= \frac{d}{r^2} \cos \theta$$

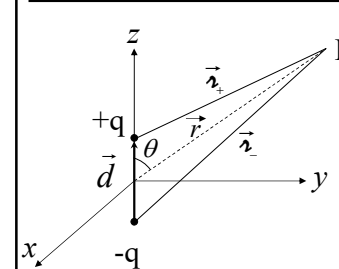
$$\rightarrow V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{qd \cos \theta}{r^2}$$

But $d \cos \theta = \vec{d} \cdot \hat{r}$, $\vec{d} = d\hat{z}$

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{q\vec{d} \cdot \hat{r}}{r^2}$$

3

Simplest possible charge system Potential



$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{q\vec{d} \cdot \hat{r}}{r^2}$$

$$\vec{p} = q\vec{d} \quad \text{Dipole Moment}$$

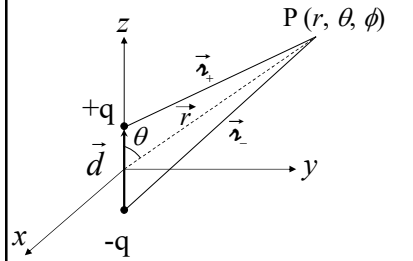
$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot \hat{r}}{r^2}$$

or

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot \vec{r}}{r^3}$$

4

Simplest possible charge system **Electric Field**



$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{qd \cos \theta}{r^2} \quad [1]$$

$$\vec{E} = -\vec{\nabla} V$$

$$\vec{E} = -\left[\hat{r} \frac{\partial V}{\partial r} + \frac{\hat{\theta}}{r} \frac{\partial V}{\partial \theta} \right] \quad [2]$$

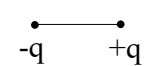
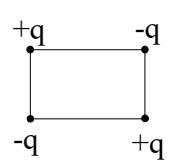
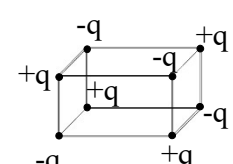
Substituting (1) into (2), gives

$$\vec{E}(r, \theta) = \frac{P}{4\pi\epsilon_0 r^3} \left[(2 \cos \theta) \hat{r} + (\sin \theta) \hat{\theta} \right] \quad [3]$$

or, in coordinate-free form $\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{1}{r^3} [3(\vec{p} \cdot \hat{r}) \hat{r} - \vec{p}]$ HW

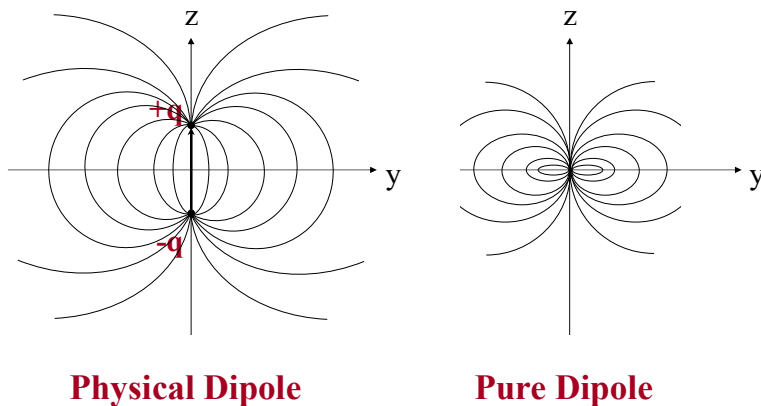
5

Important Nomenclatures

Monopole	Dipole	Quadrupole	Octopole	
				
$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	
$1/r^2$	$1/r^3$	$1/r^4$	$1/r^5$	E. Field
$1/r$	$1/r^2$	$1/r^3$	$1/r^4$	Potential

6

Just to remind you:



7

So far we have studied Electrostatics w. r. t.
individual charge/charge system/charge
distribution over a region and conductors

**Study of Electrostatics is incomplete without
considering the effect of the matter – specially
how does matter respond towards an Electric field?**

**- cause almost all applications of Electrostatics
involves field – matter interaction!**

8

Field – Matter Interaction

Matter can be solid (glass, metal, wood....), gas or liquid

→ **Conductor:** having unlimited free charged particles, not associated with the parent nucleus

→ **Insulator (DIELECTRICS):** every electron is attached to the atom/molecule.

Can only get displaced within atom/molecule

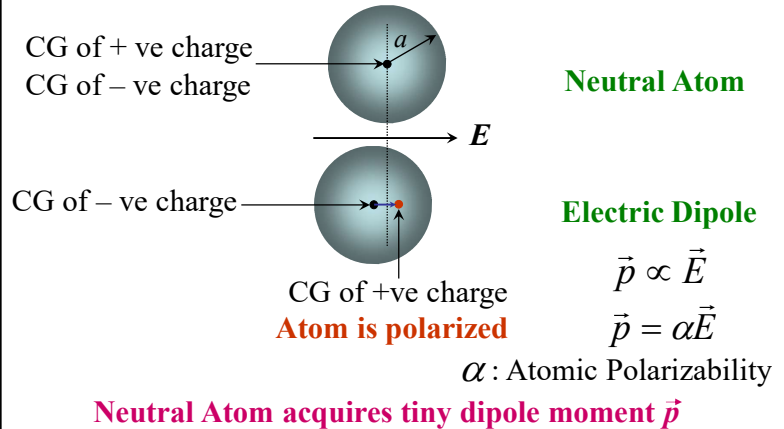
Cumulative effect of this microscopic displacement becomes extremely imp

Cause, it defines the characteristic behavior of dielectric material towards an electric field

9

Field – Matter Interaction (Case I : Atom)

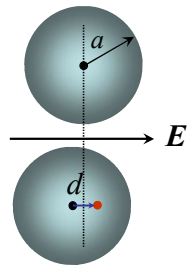
Question: What happens to a **neutral atom** when it is placed in an electric field?



10

Field – Matter Interaction (Case I : Atom)

Question: What happens to a **neutral atom** when it is placed in an electric field?



Atom is polarized

Polarization is Unidirectional

In equilibrium

$$E_{\text{ext}} = E_{\text{electron cloud}}$$

$$|\vec{E}_{\text{electron cloud}}| = \frac{1}{4\pi\epsilon_0} \frac{qd}{a^3}$$

$$|\vec{E}_{\text{ext}}| = \frac{1}{4\pi\epsilon_0} \frac{qd}{a^3} = \frac{1}{4\pi\epsilon_0} \frac{p}{a^3}$$

$$p = 4\pi\epsilon_0 a^3 E$$

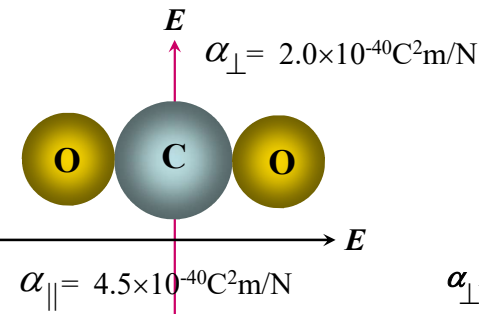
$$p = 3\epsilon_0 (4\pi a^3 / 3) E$$

$$p = (3\epsilon_0 V) E = \alpha E$$

11

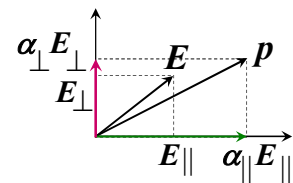
Field – Matter Interaction (Case II: Molecule)

Question: What happens to a **linear molecule** when it is placed in an electric field?



In general

$$p = \alpha_{\perp} E_{\perp} + \alpha_{\parallel} E_{\parallel}$$



12

Field – Matter Interaction (Case III: Molecule)

Question: What happens to a completely asymmetrical molecule when it is placed in an electric field?

$$\vec{p} = \alpha \vec{E}$$

In general

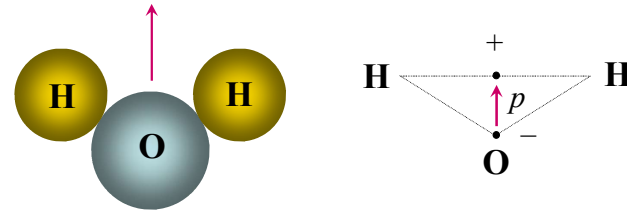
$$\begin{aligned} p_x &= \alpha_{xx} E_x + \alpha_{xy} E_y + \alpha_{xz} E_z \\ p_y &= \alpha_{yx} E_x + \alpha_{yy} E_y + \alpha_{yz} E_z \\ p_z &= \alpha_{zx} E_x + \alpha_{zy} E_y + \alpha_{zz} E_z \end{aligned} \quad \text{or} \quad \begin{pmatrix} p_x \\ p_y \\ p_z \end{pmatrix} = \begin{pmatrix} \alpha_{xx} & \alpha_{xy} & \alpha_{xz} \\ \alpha_{yx} & \alpha_{yy} & \alpha_{yz} \\ \alpha_{zx} & \alpha_{zy} & \alpha_{zz} \end{pmatrix} \begin{pmatrix} E_x \\ E_y \\ E_z \end{pmatrix}$$

α_{ij} : Polarizability Tensor

Polarization is not linear

13

Field – Matter Interaction (Case IV: Polar Molecule)

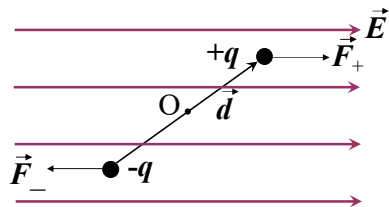


Polar Molecule

14

Field – Matter Interaction (Case IV: Polar Molecule)

Question: polar molecule in uniform electric field



Net Force = 0

Net Torque (about the center of the dipole)

$$\begin{aligned} \vec{N} &= \vec{r}_+ \times \vec{F}_+ + \vec{r}_- \times \vec{F}_- \\ &= (\vec{d}/2) \times (q\vec{E}) + (-\vec{d}/2) \times (-q\vec{E}) = q\vec{d} \times \vec{E} \end{aligned}$$

$$\vec{N} = \vec{p} \times \vec{E}$$

15

Polarization:

Question: What happens to a dielectric material when it is placed in an electric field?

If made of neutral atoms/non-polar molecules, **field induces a tiny dipole moment** to each, in its own direction

If made of polar molecules, each permanent dipole will **experience torque** and align themselves in the direction of the field



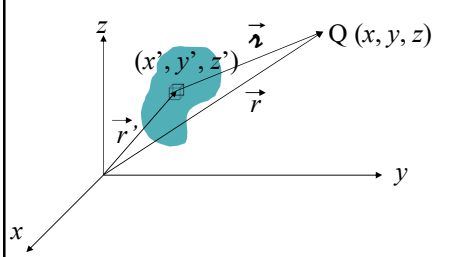
Lots of tiny dipoles pointing in the direction of the field

Material is polarized

dipole moment per unit volume = POLARIZATION ($\vec{P}$)

16

Field of a Polarized Dielectric



$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot \hat{n}}{r^2}$$

$$\vec{p} = \vec{P}(\vec{r}') d\tau'$$

$$dV(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{\hat{n} \cdot \vec{P}(\vec{r}')}{r^2} d\tau'$$

$$V(\vec{r}) = \int dV(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int_V \frac{\hat{n} \cdot \vec{P}(\vec{r}')}{r^2} d\tau'$$

$$r^2 = (x-x')^2 + (y-y')^2 + (z-z')^2$$

$$\vec{\nabla}' \left(\frac{1}{r} \right) = \frac{\hat{n}}{r^2} \quad \Rightarrow \quad V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int_V \vec{P} \cdot \vec{\nabla}' \left(\frac{1}{r} \right) d\tau'$$

17

Field of a Polarized Dielectric

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int_V \vec{P} \cdot \vec{\nabla}' \left(\frac{1}{r} \right) d\tau'$$

$$\vec{\nabla}' \cdot \left(\frac{\vec{P}}{r} \right) = \frac{1}{r} (\vec{\nabla}' \cdot \vec{P}) + \vec{P} \cdot \vec{\nabla}' \left(\frac{1}{r} \right)$$

$$\vec{P} \cdot \vec{\nabla}' \left(\frac{1}{r} \right) = \vec{\nabla}' \cdot \left(\frac{\vec{P}}{r} \right) - \frac{1}{r} (\vec{\nabla}' \cdot \vec{P})$$

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int_V \vec{\nabla}' \cdot \left(\frac{\vec{P}}{r} \right) d\tau' - \int_V \frac{1}{r} (\vec{\nabla}' \cdot \vec{P}) d\tau'$$

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \oint_S \left(\frac{\vec{P}}{r} \right) \cdot d\vec{a}' - \int_V \frac{1}{r} (\vec{\nabla}' \cdot \vec{P}) d\tau'$$

18

Field of a Polarized Dielectric

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \oint_S \left(\frac{\vec{P}}{r} \right) \cdot d\vec{a}' - \int_V \frac{1}{r} (\vec{\nabla}' \cdot \vec{P}) d\tau'$$

Potential due to
surface charge
distribution

Potential due to
volume charge
distribution

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \oint_S \left(\frac{\sigma_b}{r} \right) da' + \int_V \frac{\rho_b}{r} d\tau'$$

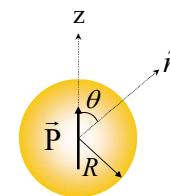
→ $\sigma_b = \vec{P} \cdot \hat{n}$ **Bound surface charge density**

→ $\rho_b = -\vec{\nabla} \cdot \vec{P}$ **Bound volume charge density**

19

Example

$\vec{E}$ produced by a uniformly polarized sphere of radius R



$$\rho_b = -\vec{\nabla} \cdot \vec{P} = 0 \quad \text{cause } \vec{P} \text{ is uniform}$$

$$\sigma_b = \vec{P} \cdot \hat{n} = P \cos \theta$$

Field by σ_b plastered over the surface of a sphere

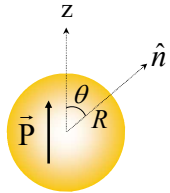
Borrowing the result from
BVP in spherical polar CS
(Ex. 3.9)

$$V(r, \theta) = \begin{cases} \frac{P}{3\epsilon_0} r \cos \theta & r \leq R \\ \frac{P}{3\epsilon_0} \frac{R^3}{r^2} \cos \theta & r \geq R \end{cases}$$

20

Example

$\vec{E}$ produced by a uniformly polarized sphere of radius R



$$V(r, \theta) = \begin{cases} \frac{P}{3\epsilon_0} r \cos \theta & r \leq R \\ \frac{P}{3\epsilon_0} \frac{R^3}{r^2} \cos \theta & r \geq R \end{cases}$$

$r < R$

$$r \cos \theta = z$$

$$V = \frac{P}{3\epsilon_0} r \cos \theta = \frac{P}{3\epsilon_0} z$$

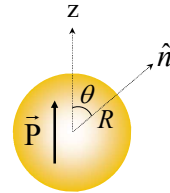
$$\rightarrow \vec{E} = -\vec{\nabla} V = -\frac{P}{3\epsilon_0} \hat{z} = -\frac{1}{3\epsilon_0} \vec{P}$$

Field is constant inside

21

Example

$\vec{E}$ produced by a uniformly polarized sphere of radius R



$$V(r, \theta) = \begin{cases} \frac{P}{3\epsilon_0} r \cos \theta & r \leq R \\ \frac{P}{3\epsilon_0} \frac{R^3}{r^2} \cos \theta & r \geq R \end{cases}$$

$r > R$

$$V = \frac{P}{3\epsilon_0} \frac{R^3}{r^2} \cos \theta \quad \text{but} \quad P = \frac{p}{(4\pi R^3)/3}$$

$$\rightarrow V = \frac{p}{3\epsilon_0 [4\pi R^3/3]} \frac{R^3}{r^2} \cos \theta = \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot \hat{r}}{r^2}$$

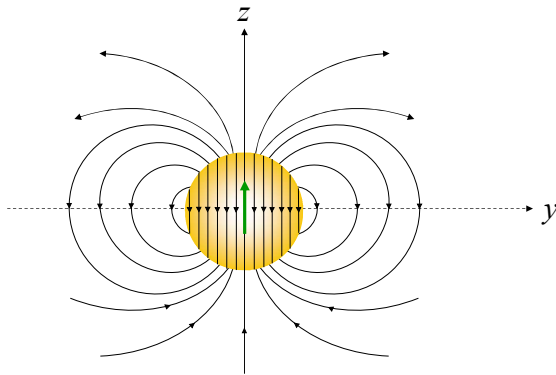
Potential of a perfect dipole at the origin

Field outside is of the dipole kept at origin

22

Example

$\vec{E}$ produced by a uniformly polarized sphere of radius R

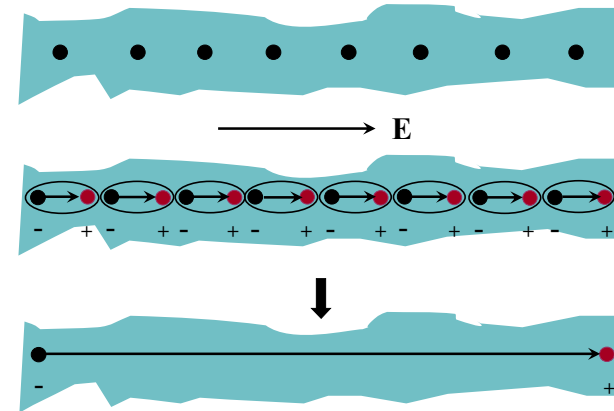


Dipole moment of the sphere $\vec{p} = \frac{4\pi R^3}{3} \vec{P}$

23

Physics behind the Bound Charge/Charge Density

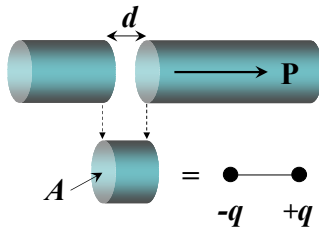
Uniform Polarization



24

Physics behind the Bound Charge/Charge Density

Uniform Polarization



$$p = P(Ad)$$

$$p = qd$$

$$q = PA$$

$$\frac{q}{A} = P = \sigma_b$$

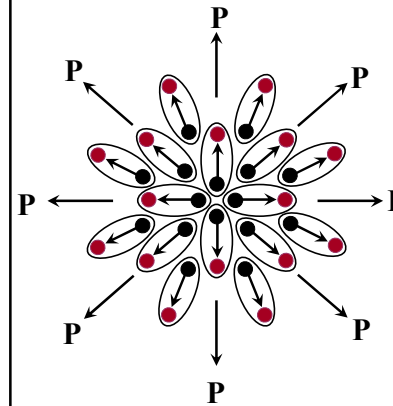
$$\sigma_b = \frac{q}{A_{\text{end}}} = P \cos \theta = \vec{P} \cdot \hat{n}$$

$$A = A_{\text{end}} \cos \theta$$

25

Physics behind the Bound Charge/Charge Density

Non-uniform Polarization



Divergence of $\vec{P}$ results in an accumulation of -ve charge at the center

Essentially the net bound charge piling up throughout the volume **is equal and opposite** to the amount that has been pushed out from the surface.

$$\int_V \rho_b d\tau = -\oint_S \vec{P} \cdot d\vec{a} = -\int_V (\vec{\nabla} \cdot \vec{P}) d\tau$$

$$\rho_b = -\vec{\nabla} \cdot \vec{P}$$

26

Gauss's Law in the Presence of Dielectric

Effect of Polarization is to produce accumulation of bound charges $\sigma_b = \vec{P} \cdot \hat{n}$ on the surface of the dielectric

$\rho_b = -\vec{\nabla} \cdot \vec{P}$ within the dielectric

Field due to polarization of the medium is just the field of these bound charges

Lets put together the cumulative effect of entire charges (bound + free charges)

Total charge density within the dielectric is $\rho = \rho_b + \rho_f$

$$\epsilon_0 \vec{\nabla} \cdot \vec{E} = \rho = -\vec{\nabla} \cdot \vec{P} + \rho_f \rightarrow \vec{\nabla} \cdot (\epsilon_0 \vec{E} + \vec{P}) = \rho_f \quad \epsilon_0 \vec{E} + \vec{P} = \vec{D}$$

$$\vec{\nabla} \cdot \vec{D} = \rho_f$$

$$\int_V \vec{\nabla} \cdot \vec{D} d\tau = \int_V \rho_f d\tau$$

$$\int_S \vec{D} \cdot d\vec{a} = Q_{f \text{ enc}}$$

27

Summary

With dielectric $\vec{\nabla} \cdot \vec{D} = \rho_f$ $\vec{\nabla} \times \vec{E} = 0$

which is the macroscopic counterpart of microscopic equations

Without dielectric $\vec{\nabla} \cdot \vec{E} = \rho / \epsilon_0$ $\vec{\nabla} \times \vec{E} = 0$

as if we replaced $\rho \rightarrow \rho_f$ $\epsilon_0 \vec{E} \rightarrow \vec{D}$

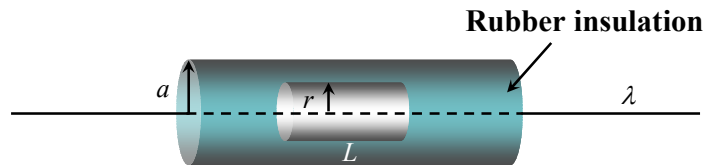
$\vec{D}$ is just like $\vec{E}$ and we can calculate it using

~~$$\vec{D}(\vec{r}) = \frac{1}{4\pi} \int_V \frac{\hat{r}}{r^2} \rho_f(\vec{r}') d\tau'$$~~

because $\epsilon_0 \vec{E} + \vec{P} = \vec{D}$ and $\vec{\nabla} \cdot \vec{D} = \rho_f$ gives only the divergence

We need to find curl. $\vec{\nabla} \times \vec{D} = \epsilon_0 (\vec{\nabla} \times \vec{E}) + (\vec{\nabla} \times \vec{P}) = (\vec{\nabla} \times \vec{P}) \neq 0$

28

Example

Find electric displacement.

$$\int_S \vec{D} \cdot d\vec{a} = Q_{f\text{enc}}$$

$$D(2\pi r L) = \lambda L$$

$$\vec{D} = \frac{\lambda}{2\pi r} \hat{r} \quad \text{valid inside/outside}$$

Outside $\vec{P} = 0$ $\epsilon_0 \vec{E} + \vec{P} = \vec{D} \rightarrow \epsilon_0 \vec{E} = \vec{D}$ $\vec{E} = \frac{\lambda}{2\pi\epsilon_0 r} \hat{r}$

Inside, we do not know $\vec{P}$, and hence can not find $\vec{E}$

29

Susceptibility, Permittivity, Dielectric constant

$$\vec{P} \propto \vec{E}$$

$$\vec{P} = \epsilon_0 \chi_e \vec{E}$$

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P} = \epsilon_0 \vec{E} + \epsilon_0 \chi_e \vec{E} = \epsilon_0 (1 + \chi_e) \vec{E} = \epsilon \vec{E}$$

$$\vec{D} = \epsilon \vec{E}$$

$$\epsilon = \epsilon_0 (1 + \chi_e)$$

In vacuum, $\chi_e = 0 \rightarrow \epsilon = \epsilon_0$

In conductor, $\chi_e \rightarrow \infty$

$$\epsilon_r = \frac{\epsilon}{\epsilon_0} = (1 + \chi_e)$$

Relative permittivity/dielectric constt

30

Few Important points

$\epsilon \neq \epsilon(\vec{r})$	Homogeneous Medium
$\epsilon = \epsilon(\vec{r})$	Inhomogeneous Medium
$\epsilon \neq \epsilon(\vec{r}, \vec{E})$ $\vec{P} = \epsilon_0 \chi_e \vec{E}$	
$\vec{D} = \epsilon \vec{E}$	Linear Medium
$\epsilon = \epsilon(\vec{E})$ $\vec{P} = \epsilon_0 \chi_e \vec{E}$	
$\vec{D} = \epsilon \vec{E}$	Nonlinear Medium
$\epsilon = \text{Constant}$	
$\vec{D} \parallel \vec{E}$ $\vec{P} \parallel \vec{E}$	Isotropic Medium
$\epsilon = \text{Second rank tensor}$	
$\vec{D} \parallel \vec{E}$ $\vec{P} \parallel \vec{E}$	Non-isotropic Medium

31

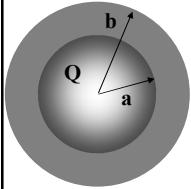
Few Important points

$\epsilon \neq \epsilon(\vec{r})$	Homogeneous Medium
$\epsilon \neq \epsilon(\vec{r}, \vec{E})$ $\vec{P} = \epsilon_0 \chi_e \vec{E}$	
$\vec{D} = \epsilon \vec{E}$	Linear Medium
$\epsilon = \text{Constant}$	
$\vec{D} \parallel \vec{E}$ $\vec{P} \parallel \vec{E}$	Isotropic Medium

32

Example

**Metal sphere of radius a carries a charge Q .
is surrounded out to a radius b , by a linear dielectric ϵ .
Potential at the center?**



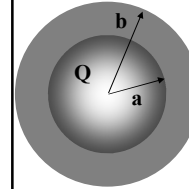
To compute $V \rightarrow$ need $\vec{E}$
To compute $\vec{E} \rightarrow$ need bound charge
To compute bound charge $\rightarrow$ need $\vec{P}$
To compute $\vec{P} \rightarrow$ need $\vec{E}$?

What we have: *free charge* Q } Calculate $\vec{D}$
: spherical symmetry

33

Example

**Metal sphere of radius a carries a charge Q .
is surrounded out to a radius b , by a linear dielectric ϵ .
Potential at the center?**



For $r > a$ $\oint \vec{D} \cdot d\vec{a} = Q_{f \text{ enc}}$

$$\vec{D} = \frac{Q}{4\pi r^2} \hat{r}$$

Inside the metal sphere, $\vec{E} = \vec{P} = \vec{D} = 0$

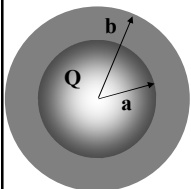
As $\vec{D} = \epsilon \vec{E}$

$$\vec{E} = \begin{cases} \frac{Q}{4\pi \epsilon r^2} \hat{r} & \text{for } a < r < b \\ \frac{Q}{4\pi \epsilon_0 r^2} \hat{r} & \text{for } r > b \end{cases}$$

34

Example

**Metal sphere of radius a carries a charge Q .
is surrounded out to a radius b , by a linear dielectric ϵ .
Potential at the center?**



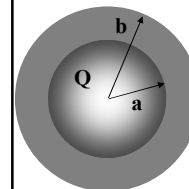
Potential at the center is

$$\begin{aligned} V &= -\int_{\infty}^0 \vec{E} \cdot d\vec{l} \\ &= -\int_{\infty}^b \left(\frac{Q}{4\pi \epsilon_0 r^2} \right) dr - \int_b^a \left(\frac{Q}{4\pi \epsilon r^2} \right) dr - \int_a^0 (0) dr \\ &= \frac{Q}{4\pi} \left(\frac{1}{\epsilon_0 b} + \frac{1}{\epsilon a} + \frac{1}{\epsilon b} \right) \end{aligned}$$

35

Example

**Metal sphere of radius a carries a charge Q .
is surrounded out to a radius b , by a linear dielectric ϵ .
Potential at the center?**



Potential at the center is

$$V = \frac{Q}{4\pi} \left(\frac{1}{\epsilon_0 b} + \frac{1}{\epsilon a} + \frac{1}{\epsilon b} \right)$$

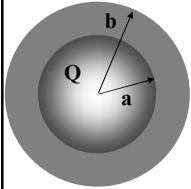
we got it without computing $\vec{P}$ or bound charges

Lets now compute $\vec{P}$ and the bound charges

36

Example

**Metal sphere of radius a carries a charge Q .
is surrounded out to a radius b , by a linear dielectric ϵ .
 $\vec{P}$? Must essentially be defined in the dielectric region**



$$\vec{P} = \epsilon_0 \chi_e \vec{E}$$

But $\vec{E} = \frac{Q}{4\pi\epsilon r^2} \hat{r}$ for $a < r < b$

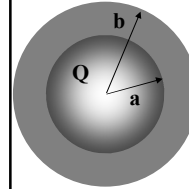
Hence, $\vec{P} = \epsilon_0 \chi_e \vec{E} = \frac{\epsilon_0 \chi_e Q}{4\pi\epsilon r^2} \hat{r}$

But how about bound charges?

37

Example

**Metal sphere of radius a carries a charge Q .
is surrounded out to a radius b , by a linear dielectric ϵ .
Bound charge densities:**



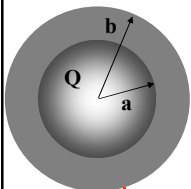
$$\vec{P} = \frac{\epsilon_0 \chi_e Q}{4\pi\epsilon r^2} \hat{r}$$

$$\rho_b = -\vec{\nabla} \cdot \vec{P} = 0 \quad \because \quad \vec{\nabla} \cdot \left(\frac{\hat{r}}{r^2} \right) = 0$$

38

Example

**Metal sphere of radius a carries a charge Q .
is surrounded out to a radius b , by a linear dielectric ϵ .
Bound charge densities:**



$$\vec{P} = \frac{\epsilon_0 \chi_e Q}{4\pi\epsilon r^2} \hat{r}$$

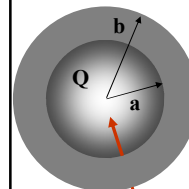
$$\rho_b = -\vec{\nabla} \cdot \vec{P} = 0$$

$$\sigma_b = \vec{P} \cdot \hat{n} = \begin{cases} \frac{\epsilon_0 \chi_e Q}{4\pi\epsilon b^2} & \text{at the outer surface} \end{cases}$$

39

Example

**Metal sphere of radius a carries a charge Q .
is surrounded out to a radius b , by a linear dielectric ϵ .
Bound charge densities:**



$$\vec{P} = \frac{\epsilon_0 \chi_e Q}{4\pi\epsilon r^2} \hat{r}$$

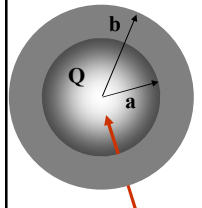
$$\rho_b = -\vec{\nabla} \cdot \vec{P} = 0$$

$$\sigma_b = \vec{P} \cdot \hat{n} = \begin{cases} \frac{\epsilon_0 \chi_e Q}{4\pi\epsilon b^2} & \text{at the outer surface} \end{cases}$$

40

Example

**Metal sphere of radius a carries a charge Q .
is surrounded out to a radius b , by a linear dielectric ϵ .
Bound charge densities:**



$$\vec{P} = \frac{\epsilon_0 \chi_e Q}{4\pi \epsilon r^2} \hat{r}$$

$$\rho_b = -\vec{\nabla} \cdot \vec{P} = 0$$

$$\sigma_b = \vec{P} \cdot \hat{n} = \begin{cases} \frac{\epsilon_0 \chi_e Q}{4\pi \epsilon b^2} & \text{at the outer surface} \\ -\frac{\epsilon_0 \chi_e Q}{4\pi \epsilon a^2} & \text{at the inner surface} \end{cases}$$

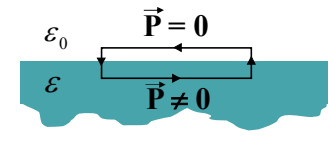
41

Question

Consider a Linear, Homogeneous, Isotropic Dielectric

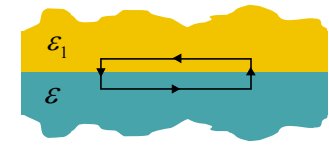
$$\vec{D} = \epsilon_0 \vec{E} + \vec{P} \quad \vec{P} = \epsilon_0 \chi_e \vec{E}$$

$$\vec{P} \text{ and } \vec{D} \propto \vec{E} \longrightarrow \vec{\nabla} \times \vec{D} = \vec{\nabla} \times \vec{P} = 0 \quad ?$$



$$\oint \vec{P} \cdot d\vec{l} \neq 0$$

$$\oint \vec{E} \cdot d\vec{l} = 0$$



$$\oint \vec{P} \cdot d\vec{l} \neq 0$$

$$\oint \vec{E} \cdot d\vec{l} = 0$$

42

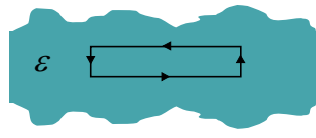
Question

Consider a Linear, Homogeneous, Isotropic Dielectric

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P} \quad \vec{P} = \epsilon_0 \chi_e \vec{E}$$

$$\vec{P} \text{ and } \vec{D} \propto \vec{E} \longrightarrow \vec{\nabla} \times \vec{D} = \vec{\nabla} \times \vec{P} = 0 \quad ?$$

However, if the region is completely filled with a linear, homogeneous, isotropic dielectric



$$\oint \vec{P} \cdot d\vec{l} = 0$$

$$\oint \vec{E} \cdot d\vec{l} = 0$$

$$\vec{\nabla} \times \vec{P} = \vec{\nabla} \times \vec{D} = 0 \longrightarrow \vec{\nabla} \cdot \vec{D} = \rho_f \quad \vec{\nabla} \times \vec{D} = 0$$

43

Question

Consider a Linear, Homogeneous, Isotropic Dielectric

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P} \quad \vec{P} = \epsilon_0 \chi_e \vec{E}$$

$$\vec{P} \text{ and } \vec{D} \propto \vec{E} \longrightarrow \vec{\nabla} \times \vec{D} = \vec{\nabla} \times \vec{P} = 0 \quad ?$$

However, if the region is completely filled with a linear, homogeneous, isotropic dielectric

$$\vec{D} = \epsilon \vec{E} \quad \epsilon = \epsilon_0 (1 + \chi_e) \quad \vec{\nabla} \cdot \vec{D} = \rho_f \rightarrow \vec{\nabla} \cdot (\epsilon \vec{E}) = \rho_f$$

$$\rightarrow \vec{\nabla} \cdot \vec{E} = \frac{\rho_f}{\epsilon} \rightarrow \oint \vec{E} \cdot d\vec{a} = \int_V \frac{\rho_f}{\epsilon} d\tau \rightarrow E 4\pi r^2 = \frac{1}{\epsilon} \int_V \rho_f d\tau$$

$$\rightarrow E = \frac{1}{4\pi \epsilon r^2} \int_V \rho_f d\tau \rightarrow E = \frac{\epsilon_0}{\epsilon} \frac{1}{4\pi \epsilon_0 r^2} \int_V \rho_f d\tau \rightarrow \vec{E} = \frac{\epsilon_0}{\epsilon} \vec{E}_{\text{vac}}$$

$$\vec{D} = \epsilon_0 \vec{E}_{\text{vac}}$$

44